

General Instructions:

1. All questions are compulsory.
2. This question paper contains 17 questions divided into five Sections A, B, C, D and E.
3. Section A comprises of 5 questions of 1 mark each, Section B comprises of 5 Questions of 2 marks each, Section C comprises of 3 Questions of 3 marks each, Section D comprises of 2 Questions of 5 marks each, and Section E comprises of 2 Case Study Based Question of 3 mark.
4. There is no overall choice.
5. Use of Calculators is not permitted.

SECTION-A(1 MARKS QUESTIONS)

Q1 The boiling points of diethyl ether, acetone and *n*-butyl alcohol are 35°C, 56°C and 118°C, respectively. Which one of the following correctly represents their boiling points in Kelvin scale?

- a. 306 K, 329 K, 391 K
- b. 308 K, 329 K, 392 K
- c. 308 K, 329 K, 391 K
- d. 329 K, 392 K, 308 K

Q2 Select the correct statement(s) regarding nucleus.

- (i) It has a double-layered covering.
 - (ii) It contains chromosomes.
 - (iii) It helps in the manufacture of lipids.
 - (iv) It plays a central role in cellular reproduction.
- a. Only (iii)
 - b. (i) and (ii)
 - c. (ii) and (iv)
 - d. (i), (ii) and (iv)

MINOR TEST-4
Max Marks =40

SCIENCE

CLASS 9TH
Set-B

Q3 The velocity-time graph of a particle is not a straight line. Its acceleration is:

- a. zero b. constant c. negative d. variable

Q4 Find the average speed of a bicycle if it completes two round of a circular track of radius 140 m twice in 5 min 52 sec.

- a. 10 m/s b. 5 m/s c. 2 m/s d. 4 m/s

Q5 Which of these is not related to endoplasmic reticulum?

- a. It behaves as transport channel for proteins between nucleus and cytoplasm.
b. It transports materials between various regions in cytoplasm.
c. It can be the site of energy generation.
d. It can be the site for some biochemical activities of the cell.

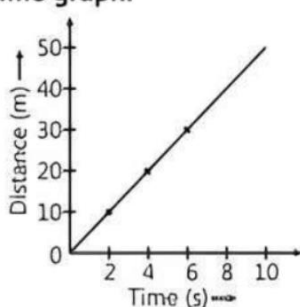
SECTION -B (2 MARKS QUESTIONS)

Q6 On a hot sunny day, why do people sprinkle water on the roof or open ground?

Q7 An object *P* is moving with a constant velocity for 5 minutes. Another object *Q* is moving with changing velocity for 5 minutes. Out of these two objects, which one possesses acceleration? Explain.

Q8

Make a velocity-time graph from the following distance-time graph.



Q9 Differentiate between diffusion and osmosis.

Q10 Why is plasma membrane also called selectively permeable membrane ?

SECTION -C (3 MARKS QUESTIONS)

Q11 Differentiate between Evaporation and Boiling

Q12 What are the main functions of each of the following cell components? (Any three)

(i) Plasma membrane

(ii) Chromosomes

(iii) Lysosomes

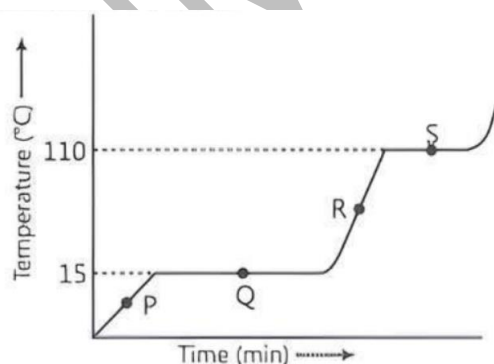
(iv) Ribosomes

(v) Nucleus

Q13 Draw a neat labelled diagram of a Animal Cell.

SECTION – D(5 MARKS QUESTIONS)

Q14 The graph shows the heating curve for a pure substance. The temperature rises with time as the substance is heated.

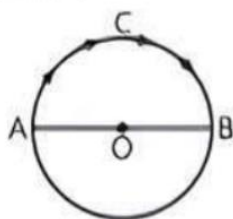


(i) What is the physical state of the substance at the points **P, Q, R and S**?

(ii) What is the melting point of the substance?

- (iii) What is its boiling point?
- (iv) What happens to the temperature while the substance is changing state?
- (v) The substance is not water. How can you judge from the graph?

Q15 An insect moves along a circular path of radius 10 cm with a constant speed. It takes 1 min to move from a point on the path to the diametrically opposite point. Find:



- (i) the distance covered
- (ii) the speed
- (iii) the displacement
- (iv) the average velocity

SECTION- E (3 MARKS QUESTIONS)

Q16 A train travels from one station to the next. The driver of train A starts from rest at time $t = 0$ and accelerates uniformly for the first 20 s. At time $t = 20$ s, train reaches its top speed of 25 ms^{-1} , then travels at this speed for further 30 s before decelerating uniformly to rest. Total time for the journey of train A is 60 s. Another train B is travelling in the parallel of train A with zero initial speed at $t = 0$ and then accelerates uniformly for first 10 s. At time $t = 10$ s it reaches its top speed of 30 ms^{-1} , then travels at this speed for further 20 s, before decelerating uniformly to rest. Total time for the journey of train B is 80 s.

Read the paragraph and answer the questions:

- a) What is the deceleration of the train A as it comes to rest?
- b) In which time interval, speed of train B is constant?
- c) What is the initial speed of trains A and B?

Q17 Vasu was helping his mother in laying the table when they had some guest for dinner. Vasu was about to sprinkle salt on salad for dressing. His mother stopped him from doing so and told him that it is too early to sprinkle salt on the salad, he should do so only when they are seated for having the dinner.

- (a) What would happen if salt is sprinkled on the salad?
- (b) Which property of cells is seen in adding salt to it?
- (c) What is membrane biogenesis?

AIMERS CLASSES